

Отчёт к лабораторной работе №2 по предмету «Администрирование в ОС Linux» «Управление разделами дисковой системы, создание и монтирование файловых систем»

4 диска:

```
[02-03-2026 12:52:10] {66381ca2} student@server:~$ hostname
server
[02-03-2026 12:52:22] {66381ca2} student@server:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 15G 0 disk
├─sda1 8:1 0 1M 0 part
└─sda2 8:2 0 15G 0 part /
sdb 8:16 0 2G 0 disk
sdc 8:32 0 2G 0 disk
sdd 8:48 0 2G 0 disk
sde 8:64 0 2G 0 disk
sr0 11:0 1 1024M 0 rom
```

1) Создан раздел /dev/sdb1 размером 500 мб на первом добавленном диске /dev/sdb, начало первый свободный сектор

```
[02-03-2026 18:17:38] {40ab33fa} student@server:~$ sudo fdisk /dev/sdb
Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-4194303, default 2048):
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-4194303, default 4194303): +500M
Created a new partition 1 of type 'Linux' and of size 500 MiB.
Partition #1 contains a ext4 signature.

Do you want to remove the signature? [Y]es/[N]o: Y
The signature will be removed by a write command.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[02-03-2026 18:48:58] {40ab33fa} student@server:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 15G 0 disk
├─sda1 8:1 0 1M 0 part
└─sda2 8:2 0 15G 0 part /
sdb 8:16 0 2G 0 disk
├─sdb1 8:17 0 500M 0 part
└─sdb 8:16 0 2G 0 disk
sdc 8:32 0 2G 0 disk
sdd 8:48 0 2G 0 disk
sde 8:64 0 2G 0 disk
sr0 11:0 1 1024M 0 rom
```

2) Получен идентификатор созданного раздела /dev/sdb1 и сохранён в файл /root/sdb1_uuid.txt

```
[02-03-2026 18:51:56] {40ab33fa} student@server:~$ sudo blkid -s PARTUUID -o value /dev/sdb1 | sudo tee /root/sdb1_uuid.txt >/dev/null
[02-03-2026 18:52:04] {40ab33fa} student@server:~$ sudo cat /root/sdb1_uuid.txt
18db5f93-01
[02-03-2026 18:52:11] {40ab33fa} student@server:~$ sudo ls -l /root/sdb1_uuid.txt
-rw-r--r-- 1 root root 12 Mar  2 18:52 /root/sdb1_uuid.txt
```

3) На разделе /dev/sdb1 создана файловая система ext4 с размером блока 4096 байт

```
[02-03-2026 18:52:20] {40ab33fa} student@server:~$ sudo mkfs.ext4 -b 4096 /dev/sdb1
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 128000 4k blocks and 128000 inodes
Filesystem UUID: 38737721-0302-4d70-b8cd-e82a21fbc00d
Superblock backups stored on blocks:
    32768, 98304

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

[02-03-2026 18:54:39] {40ab33fa} student@server:~$ sudo blkid /dev/sdb1
/dev/sdb1: UUID="38737721-0302-4d70-b8cd-e82a21fbc00d" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="18db5f93-01"
[02-03-2026 18:54:47] {40ab33fa} student@server:~$ sudo dumpe2fs -h /dev/sdb1 | grep -E "Block size|Filesystem UUID|Filesystem volume name"
dumpe2fs 1.47.0 (5-Feb-2023)
Filesystem volume name: <none>
Filesystem UUID: 38737721-0302-4d70-b8cd-e82a21fbc00d
Block size: 4096
```

4) Выведены текущие параметры суперблока файловой системы /dev/sdb1

```
[02-03-2026 19:05:17] {40ab33fa} student@server:~$ sudo dumpe2fs -h /dev/sdb1
dumpe2fs 1.47.0 (5-Feb-2023)
Filesystem volume name: <none>
Last mounted on: <not available>
Filesystem UUID: 38737721-0302-4d70-b8cd-e82a21fbc00d
Filesystem magic number: 0xEF53
Filesystem revision #: 1 (dynamic)
Filesystem features: has_journal ext_attr resize_inode dir_index filetype extent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Filesystem flags: signed_directory_hash
Default mount options: user_xattr acl
Filesystem state: clean
Errors behavior: Continue
Filesystem OS type: Linux
Inode count: 128000
Block count: 128000
Reserved block count: 6400
Overhead clusters: 12296
Free blocks: 115698
Free inodes: 127989
First block: 0
Block size: 4096
Fragment size: 4096
Group descriptor size: 64
Reserved GDT blocks: 62
Blocks per group: 32768
Fragments per group: 32768
Inodes per group: 32000
Inode blocks per group: 2800
Flex block group size: 16
Filesystem created: Mon Mar 2 18:54:39 2026
Last mount time: n/a
Last write time: Mon Mar 2 18:54:39 2026
Mount count: 0
Maximum mount count: -1
Last checked: Mon Mar 2 18:54:39 2026
Check interval: 0 (<none>)
Lifetime writes: 273 kB
Reserved blocks uid: 0 (user root)
Reserved blocks gid: 0 (group root)
First inode: 11
Inode size: 256
Required extra isize: 32
Desired extra isize: 32
Journal inode: 8
Default directory hash: half_md4
Directory Hash Seed: b6a33ac-8ba4-47ec-97b0-cc3b073921ba
Journal backup: inode blocks
Checksum type: crc32c
Checksum: 0x01465c28
Journal features: (none)
Total journal size: 16M
Total journal blocks: 4096
Max transaction length: 4096
Fast commit length: 0
Journal sequence: 0x00000001
Journal start: 0
```

```
[02-03-2026 19:07:20] {40ab33fa} student@server:~$ sudo tune2fs -l /dev/sdb1
tune2fs 1.47.0 (5-Feb-2023)
Filesystem volume name: <none>
Last mounted on: <not available>
Filesystem UUID: 38737721-0302-4d70-b8cd-e82a21fbc00d
Filesystem magic number: 0xEF53
Filesystem revision #: 1 (dynamic)
Filesystem features: has_journal ext_attr resize_inode dir_index filetype extent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Filesystem flags: signed_directory_hash
Default mount options: user_xattr acl
Filesystem state: clean
Errors behavior: Continue
Filesystem OS type: Linux
Inode count: 128000
Block count: 128000
Reserved block count: 6400
Overhead clusters: 12296
Free blocks: 115698
Free inodes: 127989
First block: 0
Block size: 4096
Fragment size: 4096
Group descriptor size: 64
Reserved GDT blocks: 62
Blocks per group: 32768
Fragments per group: 32768
Inodes per group: 32000
Inode blocks per group: 2800
Flex block group size: 16
Filesystem created: Mon Mar 2 18:54:39 2026
Last mount time: n/a
Last write time: Mon Mar 2 18:54:39 2026
Mount count: 0
Maximum mount count: -1
Last checked: Mon Mar 2 18:54:39 2026
Check interval: 0 (<none>)
Lifetime writes: 273 kB
Reserved blocks uid: 0 (user root)
Reserved blocks gid: 0 (group root)
First inode: 11
Inode size: 256
Required extra isize: 32
Desired extra isize: 32
Journal inode: 8
Default directory hash: half_md4
Directory Hash Seed: b6a33ac-8ba4-47ec-97b0-cc3b073921ba
Journal backup: inode blocks
Checksum type: crc32c
Checksum: 0x01465c28
```

5) Для /dev/sdb1 настроен автоматический запуск проверки ФС через 2 месяца или каждое второе монтирование

```
[02-03-2026 18:57:48] {40ab33fa} student@server:~$ sudo tune2fs -c 2 -i 2m /dev/sdb1
tune2fs 1.47.0 (5-Feb-2023)
Setting maximal mount count to 2
Setting interval between checks to 5184000 seconds
[02-03-2026 19:02:43] {40ab33fa} student@server:~$ sudo tune2fs -l /dev/sdb1 | grep -E "Mount count|Maximum mount count|Check interval|Last checked|Next check after"
Mount count: 0
Maximum mount count: 2
Last checked: Mon Mar 2 18:54:39 2026
Check interval: 5184000 (2 months)
Next check after: Fri May 1 18:54:39 2026
```

6) Создан каталог /mnt/newdisk, файловая система /dev/sdb1 смонтирована в него

```
[02-03-2026 19:02:51] {40ab33fa} student@server:~$ sudo mkdir -p /mnt/newdisk
[02-03-2026 19:07:18] {40ab33fa} student@server:~$ sudo mount /dev/sdb1 /mnt/newdisk
[02-03-2026 19:07:24] {40ab33fa} student@server:~$ mount | grep /mnt/newdisk
/dev/sdb1 on /mnt/newdisk type ext4 (rw,relatime)
[02-03-2026 19:07:29] {40ab33fa} student@server:~$ lsblk -f
NAME        FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
└─sda1
└─sda2 ext4  1.0          e8346619-430d-4c28-bbd9-be585fd5d98a    10.4G    24% /
sdb
└─sdb1 ext4  1.0          38737721-0302-4d70-b8cd-e82a21fbc00d    416.9M     0% /mnt/newdisk
sdc
sdd
sde
sr0
```

7) В домашнем каталоге пользователя root создана символическая ссылка /root/newdisk_link на смонтированную файловую систему /mnt/newdisk

```

[02-03-2026 19:07:34] {40ab33fa} student@server:~$ sudo ln -s /mnt/newdisk /root/newdisk_link
[02-03-2026 19:09:54] {40ab33fa} student@server:~$ sudo ls -l /root/newdisk_link
lrwxrwxrwx 1 root root 12 Mar 2 19:09 /root/newdisk_link -> /mnt/newdisk
[02-03-2026 19:09:59] {40ab33fa} student@server:~$ sudo ls -ld /root/newdisk_link/
dwxr-xr-x 3 root root 4096 Mar 2 18:54 /root/newdisk_link/

```

8) В смонтированной файловой системе /mnt/newdisk создан каталог /mnt/newdisk/lab2_data

```
[02-03-2026 19:10:04] {40ab33fa} student@server:~$ sudo mkdir -p /mnt/newdisk/lab2_data
[sudo] password for student:
[02-03-2026 19:33:23] {40ab33fa} student@server:~$ sudo ls -ld /mnt/newdisk/lab2_data
drwxr-xr-x 2 root root 4096 Mar  2 19:33 /mnt/newdisk/lab2_data
[02-03-2026 19:33:31] {40ab33fa} student@server:~$ df -h /mnt/newdisk
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1        452M   28K  417M   1% /mnt/newdisk
```

9) Настроено автоматическое монтирование /dev/sdb1 в /mnt/newdisk через /etc/fstab с параметрами noexec,noatime

```
[02-03-2026 19:33:37] {40ab33fa} student@server:~$ sudo cp /etc/fstab /etc/fstab.bak_lab2
[02-03-2026 19:37:31] {40ab33fa} student@server:~$ UUID_SDB1=$(sudo blkid -s UUID -o value /dev/sdb1)
[02-03-2026 19:37:42] {40ab33fa} student@server:~$ echo "UUID=${UUID_SDB1} /mnt/newdisk ext4 defaults,noexec,noatime 0 2" | sudo tee -a /etc/fstab
UUID=38737721-0302-4d70-b8cd-e82a21fbc00d /mnt/newdisk ext4 defaults,noexec,noatime 0 2
[02-03-2026 19:37:47] {40ab33fa} student@server:~$ sudo mount -a
[02-03-2026 19:37:52] {40ab33fa} student@server:~$ mount | grep /mnt/newdisk
/dev/sdb1 on /mnt/newdisk type ext4 (rw,relatime)
[02-03-2026 19:37:57] {40ab33fa} student@server:~$ sudo reboot

Broadcast message from root@server on pts/1 (Mon 2026-03-02 19:38:04 MSK):

The system will reboot now!

[02-03-2026 19:38:04] {40ab33fa} student@server:~$ Connection to localhost closed by remote host.
Connection to localhost closed.
[02-03-2026 20:21:19] {66381ca2} student@server:~$ mount | grep /mnt/newdisk
/dev/sdb1 on /mnt/newdisk type ext4 (rw,noexec,noatime)
[03-03-2026 00:15:37] {66381ca2} student@server:~$ cat /etc/fstab | grep /mnt/newdisk
UUID=38737721-0302-4d70-b8cd-e82a21fbc00d /mnt/newdisk ext4 defaults,noexec,noatime 0 2
[03-03-2026 00:15:43] {66381ca2} student@server:~$ echo -e '#!/bin/bash/necho test_noexec' | sudo tee /mnt/newdisk/test_noexec.sh >/dev/null
[sudo] password for student:
[03-03-2026 00:15:56] {66381ca2} student@server:~$ sudo chmod +x /mnt/newdisk/test_noexec.sh
[03-03-2026 00:16:05] {66381ca2} student@server:~$ /mnt/newdisk/test_noexec.sh
-bash: /mnt/newdisk/test_noexec.sh: Permission denied
```

10) Раздел `/dev/sdb1` увеличен до 1 Гб с сохранением начального сектора, затем файловая система `ext4` расширена

```
[[03-03-2026 00:16:10] 66381ca2] 66381ca2] student@server:~$ sudo umount /mnt/newdisk
[[03-03-2026 00:22:04] 66381ca2] student@server:~$ sudo fdisk /dev/sdb

Welcome to fdisk (util-linux 2.39.3).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): p
Disk /dev/sdb: 2 GiB, 2147483648 bytes, 4194304 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x18db5f93


Device      Boot Start      End Sectors  Size Id Type
/dev/sdb1             2048 1026047 1024000  500M 83 Linux


Command (m for help): d
Selected partition 1
partition 1 has been deleted.


Command (m for help): 1
1: unknown command


Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-4194303, default 2048): 2048
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-4194303, default 4194303): +1G

Created a new partition 1 of type 'Linux' and of size 1 GiB.
Partition #1 contains a ext4 signature.


Do you want to remove the signature? [Y]es/[N]o: N


Command (m for help): w

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

```
[[03-03-2026 00:24:01] {66381ca2} student@server:~$ sudo partprobe /dev/sdb || sudo reboot
[[03-03-2026 00:24:44] {66381ca2} student@server:~$ sudo e2fsck -f /dev/sdb1
e2fsck 1.47.0 (5-Feb-2023)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/sdb1: 13/128000 files (0.0% non-contiguous), 12304/128000 blocks
[[03-03-2026 00:24:51] {66381ca2} student@server:~$ sudo resizefs /dev/sdb1
resize2fs 1.47.0 (5-Feb-2023)
Resizing the filesystem on /dev/sdb1 to 262144 (4k) blocks.
The filesystem on /dev/sdb1 is now 262144 (4k) blocks long.

[[03-03-2026 00:24:56] {66381ca2} student@server:~$ sudo mount -a
NAME                FSTYPE FSVER LABEL UUID                               FSAVAIL FSUSE% MOUNTPOINTS
sdb1
├─sda1
├─sdb2 ext4 1.0      e8346619-430d-4c28-bbd9-be585fd5d98a 10.4G  24% /
sdb
├─sdb1 ext4 1.0      38737721-0302-4d70-b8cd-e82ae21fbc0d  877M   0% /mnt/newdisk
sdc
sdd
sde
srd
[[03-03-2026 00:25:09] {66381ca2} student@server:~$ df -h /mnt/newdisk
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1        925M    32K  877M   1% /mnt/newdisk
```


11) Файловая система /dev/sdb1 проверена на ошибки в безопасном режиме, после проверки раздел смонтирован обратно

```
[03-03-2026 00:25:15] {66381ca2} student@server:~$ sudo umount /mnt/newdisk
[03-03-2026 00:30:53] {66381ca2} student@server:~$ sudo e2fsck -f -n /dev/sdb1
e2fsck 1.47.0 (5-Feb-2023)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/sdb1: 13/256000 files (0.0% non-contiguous), 20440/262144 blocks
[03-03-2026 00:30:58] {66381ca2} student@server:~$ sudo mount -a
[03-03-2026 00:31:05] {66381ca2} student@server:~$ mount | grep /mnt/newdisk
/dev/sdb1 on /mnt/newdisk type ext4 (rw,noexec,noatime)
```

12) Создан раздел /dev/sdb2 размером 12 МБ и использован как внешнее журналирующее устройство для /dev/sdb1

```
[03-03-2026 00:31:09] {66381ca2} student@server:~$ sudo umount /mnt/newdisk
[03-03-2026 00:38:55] {66381ca2} student@server:~$ sudo fdisk /dev/sdb

Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help): n
Partition type
  p primary (1 primary, 0 extended, 3 free)
  e extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (2099200-4194303, default 2099200):
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2099200-4194303, default 4194303): +12M

Created a new partition 2 of type 'Linux' and of size 12 MiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[03-03-2026 00:39:19] {66381ca2} student@server:~$ sudo partprobe /dev/sdb || sudo reboot
[03-03-2026 00:39:41] {66381ca2} student@server:~$ lsblk -f
NAME        FSTYPE FSVER LABEL UUID                               FSAVAIL FSUSEN MOUNTPOINTS
sda
├─sda1
├─sda2 ext4  1.0      e8346619-430d-4c28-bbd9-be585fd5d98a  10.4G   24% /
sdb
├─sdb1 ext4  1.0      38737721-0302-4d70-b8cd-e82a21fbc00d
├─sdb2
├─sdd
├─sde
└─srf
```

```
[03-03-2026 00:40:29] {66381ca2} student@server:~$ sudo umount /mnt/newdisk
[03-03-2026 00:40:41] {66381ca2} student@server:~$ sudo tune2fs -O *has_journal /dev/sdb1
tune2fs 1.47.0 (5-Feb-2023)
[03-03-2026 00:41:51] {66381ca2} student@server:~$ sudo e2fsck -f /dev/sdb1
e2fsck 1.47.0 (5-Feb-2023)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/sdb1: 13/256000 files (0.0% non-contiguous), 16344/262144 blocks
[03-03-2026 00:44:00] {66381ca2} student@server:~$ sudo tune2fs -o device=/dev/sdb2 /dev/sdb1
tune2fs 1.47.0 (5-Feb-2023)
Creating journal on device /dev/sdb2: done
This filesystem will be automatically checked every 2 mounts or
60 days, whichever comes first. Use tune2fs -o or -l to override.
[03-03-2026 00:44:18] {66381ca2} student@server:~$ sudo e2fsck -f /dev/sdb1
e2fsck 1.47.0 (5-Feb-2023)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/sdb1: 13/256000 files (0.0% non-contiguous), 16344/262144 blocks
[03-03-2026 00:44:18] {66381ca2} student@server:~$ sudo dumpe2fs -h /dev/sdb1 | grep -E "Journal UUID|Journal inode|Filesystem features"
dumpe2fs 1.47.0 (5-Feb-2023)
Filesystem features:      has_journal ext_attr resize_inode dir_index filetype needs_recovery extent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Journal UUID:            ca97d909-58b6-4936-a97b-15e84f0e5962
```

13) На дисках /dev/sdc и /dev/sdd созданы разделы /dev/sdc1 и /dev/sdd1, занимающие весь объём и инициализированы как физические тома LVM

```
[03-03-2026 00:44:25] {66381ca2} student@server:~$ sudo fdisk /dev/sdc

Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x7e5f899c.

Command (m for help): n
Partition type
  p primary (0 primary, 0 extended, 4 free)
  e extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-4194303, default 2048):
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-4194303, default 4194303):

Created a new partition 1 of type 'Linux' and of size 2 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[03-03-2026 00:47:36] {66381ca2} student@server:~$ sudo fdisk /dev/sdd

Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x9d1b648d.

Command (m for help): n
Partition type
  p primary (0 primary, 0 extended, 4 free)
  e extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-4194303, default 2048):
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-4194303, default 4194303):

Created a new partition 1 of type 'Linux' and of size 2 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

```
[03-03-2026 00:47:54] {66381ca2} student@server:~$ sudo partprobe /dev/sdc /dev/sdd || sudo reboot
[03-03-2026 00:49:41] {66381ca2} student@server:~$ lsblk -f
NAME        FSTYPE FSVER LABEL UUID                               FSAVAIL FSUSEN MOUNTPOINTS
sda
├─sda1
├─sda2 ext4  1.0      e8346619-430d-4c28-bbd9-be585fd5d98a  10.4G   24% /
sdb
├─sdb1 ext4  1.0      38737721-0302-4d70-b8cd-e82a21fbc00d  893M    0% /mnt/newdisk
├─sdb2 jbd   1.0      ca97d909-58b6-4936-a97b-15e84f0e5962
├─sdc
├─sdd
├─sdd1
├─sde
└─srf

[03-03-2026 00:49:49] {66381ca2} student@server:~$ sudo pvcreate /dev/sdc1 /dev/sdd1
Physical volume "/dev/sdc1" successfully created.
Physical volume "/dev/sdd1" successfully created.
[03-03-2026 00:49:53] {66381ca2} student@server:~$ sudo pvs
PV          VG Fmt Attr PSize PFree
/dev/sdc1   --- ---  <2.00g <2.00g
/dev/sdd1   --- ---  <2.00g <2.00g
[03-03-2026 00:49:59] {66381ca2} student@server:~$ sudo pvdisplay
"/dev/sdc1" is a new physical volume of "2.00 GiB"
--- NEW Physical volume ---
PV Name     /dev/sdc1
VG Name
PV Size     2.00 GiB
Allocatable NO
PE Size     0
Total PE    0
Free PE     0
Allocated PE 0
PV UUID     15c6bf-c7e9-3e34-8f0f-0xaa-m6cv-WHEyYb

"/dev/sdd1" is a new physical volume of "2.00 GiB"
--- NEW Physical volume ---
PV Name     /dev/sdd1
VG Name
PV Size     2.00 GiB
Allocatable NO
PE Size     0
Total PE    0
Free PE     0
Allocated PE 0
PV UUID     fh755-2w4c-11qc-P1kq-UDbh-2skU-TIQiTa
```


14) Создана группа томов vg_lab2 на /dev/sdc1 и /dev/sdd1, создан чередующийся логический том на весь доступный объём, на нём создана файловая система ext4

```
[[03-03-2026 00:50:06] {66381ca2} student@server:~$ sudo vgcreate vg_lab2 /dev/sdc1 /dev/sdd1
Volume group "vg_lab2" successfully created
[[03-03-2026 00:54:48] {66381ca2} student@server:~$ sudo lvcreate -i 2 -I 64 -l 100%FREE -n lv_vol01 vg_lab2
Logical volume "lv_vol01" created.
[[03-03-2026 00:54:54] {66381ca2} student@server:~$ sudo mkfs.ext4 /dev/vg_lab2/lv_vol01
mkfs2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 1046528 4k blocks and 261632 inodes
Filesystem UUID: a84774bc-47e2-4be0-8337-f0db95198b23
Superblock backups stored on blocks:
32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[[03-03-2026 00:55:00] {66381ca2} student@server:~$ sudo lvs -a -o +devices
LV      VG      Attr      LSize Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert Devices
lv_vol01 vg_lab2 -wi-a----- 3.99g                                     /dev/sdc1(0),/dev/sdd1(0)
[[03-03-2026 00:55:06] {66381ca2} student@server:~$ sudo blkid /dev/vg_lab2/lv_vol01
/dev/vg_lab2/lv_vol01: UUID="a84774bc-47e2-4be0-8337-f0db95198b23" BLOCK_SIZE="4096" TYPE="ext4"
```

15) Логический том /dev/vg_lab2/lv_vol01 смонтирован в /mnt/vol01, в /etc/fstab добавлена запись автомонтирования по UUID

```
[[03-03-2026 00:55:12] {66381ca2} student@server:~$ sudo mkdir -p /mnt/vol01
[[03-03-2026 00:58:58] {66381ca2} student@server:~$ sudo mount /dev/vg_lab2/lv_vol01 /mnt/vol01
[[03-03-2026 00:59:05] {66381ca2} student@server:~$ mount | grep /mnt/vol01
/dev/mapper/vg_lab2-lv_vol01 on /mnt/vol01 type ext4 (rw,relatime,stripe=32)
[[03-03-2026 00:59:10] {66381ca2} student@server:~$ UUID_VOL01=$(sudo blkid -s UUID -o value /dev/vg_lab2/lv_vol01)
[[03-03-2026 00:59:16] {66381ca2} student@server:~$ echo "UUID=${UUID_VOL01} /mnt/vol01 ext4 defaults 0 2" | sudo tee -a /etc/fstab
UUID=a84774bc-47e2-4be0-8337-f0db95198b23 /mnt/vol01 ext4 defaults 0 2
[[03-03-2026 00:59:21] {66381ca2} student@server:~$ sudo umount /mnt/vol01
[[03-03-2026 00:59:31] {66381ca2} student@server:~$ sudo mount -a
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
[[03-03-2026 00:59:36] {66381ca2} student@server:~$ mount | grep /mnt/vol01
/dev/mapper/vg_lab2-lv_vol01 on /mnt/vol01 type ext4 (rw,relatime,stripe=32)
[[03-03-2026 00:59:42] {66381ca2} student@server:~$ df -h /mnt/vol01
Filesystem              Size  Used Avail Use% Mounted on
/dev/mapper/vg_lab2-lv_vol01 3.9G  24K  3.7G   1% /mnt/vol01
```

16) Получена информация LVM: по физическим томам (pvs, pvdisplay), группе томов (vgs, vgdisplay) и логическому тому (lvs -o +devices, lvdisplay)

```
[[03-03-2026 00:59:47] {66381ca2} student@server:~$ sudo pvs
PV      VG      Fmt Attr PSize PFree
/dev/sdc1 vg_lab2 lvm2 a-- <2.00g 0
/dev/sdd1 vg_lab2 lvm2 a-- <2.00g 0
[[03-03-2026 01:03:59] {66381ca2} student@server:~$ sudo vgs
VG      #PV #LV #SN Attr   VSize VFree
vg_lab2 2  1  0 wz--n- 3.99g 0
[[03-03-2026 01:04:04] {66381ca2} student@server:~$ sudo lvs -a -o +devices
LV      VG      Attr      LSize Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert Devices
lv_vol01 vg_lab2 -wi-a----- 3.99g                                     /dev/sdc1(0),/dev/sdd1(0)
[[03-03-2026 01:04:09] {66381ca2} student@server:~$ sudo pvdisplay
--- Physical volume ---
PV Name      /dev/sdc1
VG Name      vg_lab2
PV Size      <2.00 GiB / not usable 3.00 MiB
Allocatable  yes (but full)
PE Size      4.00 MiB
Total PE     511
Free PE      0
Allocated PE 511
PV UUID      lYEcBf-CYwy-3e34-8F0f-0xaa-m6cv-WHEVyb

--- Physical volume ---
PV Name      /dev/sdd1
VG Name      vg_lab2
PV Size      <2.00 GiB / not usable 3.00 MiB
Allocatable  yes (but full)
PE Size      4.00 MiB
Total PE     511
Free PE      0
Allocated PE 511
PV UUID      fHz7S5-2w4C-1iqC-P1k6-UDbh-2sKU-TIQtTa

[[03-03-2026 01:04:14] {66381ca2} student@server:~$ sudo vgdisplay
--- Volume group ---
VG Name      vg_lab2
System ID
Format       lvm2
Metadata Areas 2
Metadata Sequence No 2
VG Access    read/write
VG Status     resizable
MAX LV       0
Cur LV      1
Open LV      1
Max PV       0
Cur PV      2
Act PV       2
VG Size      3.99 GiB
PE Size      4.00 MiB
Total PE     1022
Alloc PE / Size 1022 / 3.99 GiB
Free PE / Size 0 / 0
VG UUID      nBZj3Q-nQ2V-ts1K-D3Gi-xV3B-QuWi-XsqHq5

[[03-03-2026 01:04:19] {66381ca2} student@server:~$ sudo lvdisplay
--- Logical volume ---
LV Path      /dev/vg_lab2/lv_vol01
LV Name      lv_vol01
VG Name      vg_lab2
LV UUID      WzTxdk-9aKk-ILNE-id1b-FZ4Q-WBM3-seCwzU
LV Write Access read/write
LV Creation host, time server, 2026-03-03 00:54:54 +0300
LV Status     available
# open       1
LV Size      3.99 GiB
Current LE   1022
Segments     1
Allocation   inherit
Read ahead sectors auto
- currently set to 512
Block device 252:0
```

17) Дополнительный диск /dev/sde1 добавлен в VG vg_lab2, логический том lv_vol01 расширен на 100% нового диска

```
[03-03-2026 01:04:24] {66381ca2} student@server:~$ sudo fdisk /dev/sde
Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x874141b2.

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-4194383, default 2048):
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-4194383, default 4194383):

Created a new partition 1 of type 'Linux' and of size 2 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[03-03-2026 01:30:04] {66381ca2} student@server:~$ sudo partprobe /dev/sde || sudo reboot
[03-03-2026 01:31:03] {66381ca2} student@server:~$ lsblk -f

NAME        FSTYPE     FSVER    LABEL UUID                                FSAVAIL FSUSE% MOUNTPOINTS
sda
sda1
sda2
sdb1
sdb2
sdc
sdc1
sdd
sde
sde1
sr0

[03-03-2026 01:31:07] {66381ca2} student@server:~$ sudo pvcreate /dev/sde1
Physical volume "/dev/sde1" successfully created.
[03-03-2026 01:32:53] {66381ca2} student@server:~$ sudo vgextend vg_lab2 /dev/sde1
Volume group "vg_lab2" successfully extended
[03-03-2026 01:32:49] {66381ca2} student@server:~$ sudo lvextend -l +100%PVS /dev/sde1 /dev/vg_lab2/lv_vol01
"/dev/sde1": Invalid path for Logical Volume.
Run "lvextend --help" for more information.
[03-03-2026 01:32:54] {66381ca2} student@server:~$ sudo pvs
PV          VG      Fmt  Attr PSize  PFree
/dev/sdc1   vg_lab2 lvm2  a--  <2.00g  0
/dev/sdd1   vg_lab2 lvm2  a--  <2.00g  0
/dev/sde1   vg_lab2 lvm2  a--  <2.00g  <2.00g
[03-03-2026 01:32:58] {66381ca2} student@server:~$ sudo vgs
VG          #PV #LV #SN Attr   VSize  VFree
vg_lab2     3   1   0 wz--n- <5.99g <2.00g
[03-03-2026 01:33:03] {66381ca2} student@server:~$ sudo lvs -a -o +devices
LV          VG      Attr   LSize Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert Devices
lv_vol01    vg_lab2 -wi-a- 3.99g          /dev/sdc1(0),/dev/sdd1(0)
[03-03-2026 01:35:33] {66381ca2} student@server:~$ sudo lvextend -l1 -l +100%PVS /dev/vg_lab2/lv_vol01 /dev/sde1
sudo lvs -a -o +devices
Size of logical volume vg_lab2/lv_vol01 changed from 3.99 GiB (1022 extents) to <5.99 GiB (1533 extents).
Logical volume vg_lab2/lv_vol01 successfully resized.
LV          VG      Attr   LSize Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert Devices
lv_vol01    vg_lab2 -wi-a- <5.99g          /dev/sdc1(0),/dev/sdd1(0)
lv_vol01    vg_lab2 -wi-a- <5.99g          /dev/sde1(0)
VG          #PV #LV #SN Attr   VSize  VFree
vg_lab2     3   1   0 wz--n- <5.99g 0
```

18) Файловая система ext4 на /dev/vg_lab2/lv_vol01 расширена онлайн (без отмонтирования) командой resize2fs

```
[03-03-2026 01:38:06] {66381ca2} student@server:~$ mount | grep /mnt/vol01
/dev/mapper/vg_lab2-lv_vol01 on /mnt/vol01 type ext4 (rw,relatime,stripe=32)
[03-03-2026 01:41:24] {66381ca2} student@server:~$ sudo resize2fs /dev/vg_lab2/lv_vol01
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/vg_lab2/lv_vol01 is mounted on /mnt/vol01; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 1
The filesystem on /dev/vg_lab2/lv_vol01 is now 1569792 (4k) blocks long.

[03-03-2026 01:41:30] {66381ca2} student@server:~$ df -h /mnt/vol01
Filesystem              Size  Used Avail Use% Mounted on
/dev/mapper/vg_lab2-lv_vol01 5.9G  24K  5.6G   1% /mnt/vol01
[03-03-2026 01:41:45] {66381ca2} student@server:~$ lsblk -f | grep -E "vg_lab2-lv_vol01|lv_vol01"
└─vg_lab2-lv_vol01 ext4      1.0    a84774bc-47e2-4be8-8337-f8db95198b23    5.5G  0% /mnt/vol01
└─vg_lab2-lv_vol01 ext4      1.0    a84774bc-47e2-4be8-8337-f8db95198b23    5.5G  0% /mnt/vol01
└─vg_lab2-lv_vol01 ext4      1.0    a84774bc-47e2-4be8-8337-f8db95198b23    5.5G  0% /mnt/vol01
```

19) Получена информация LVM: по физическим томам (pvs, pvdisplay), группе томов (vgs, vgdisplay) и логическому тому (lvs/lvdisplay)

```
[03-03-2026 01:41:51] {66381ca2} student@server:~$ sudo pvs
PV          VG      Fmt  Attr PSize  PFree
/dev/sdc1   vg_lab2 lvm2  a--  <2.00g  0
/dev/sdd1   vg_lab2 lvm2  a--  <2.00g  0
/dev/sde1   vg_lab2 lvm2  a--  <2.00g  0
[03-03-2026 01:45:58] {66381ca2} student@server:~$ sudo vgs
VG          #PV #LV #SN Attr   VSize  VFree
vg_lab2     3   1   0 wz--n- <5.99g <2.00g
[03-03-2026 01:45:44] {66381ca2} student@server:~$ sudo lvs -a -o +devices
LV          VG      Attr   LSize Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert Devices
lv_vol01    vg_lab2 -wi-a- <5.99g          /dev/sdc1(0),/dev/sdd1(0)
lv_vol01    vg_lab2 -wi-a- <5.99g          /dev/sde1(0)
[03-03-2026 01:45:50] {66381ca2} student@server:~$ sudo pvdisplay
--- Physical volume ---
PV Name               /dev/sdc1
VG Name               vg_lab2
PV Size               <2.00 GiB / not usable 3.00 MiB
Allocatable           yes (but full)
PE Size               4.00 MiB
Total PE              511
Free PE               0
Allocated PE          511
PV UUID               lYecBf-CYwy-3e34-8F0f-Oxaa-m6cv-MHEYyb

--- Physical volume ---
PV Name               /dev/sdd1
VG Name               vg_lab2
PV Size               <2.00 GiB / not usable 3.00 MiB
Allocatable           yes (but full)
PE Size               4.00 MiB
Total PE              511
Free PE               0
Allocated PE          511
PV UUID               fhZ7S5-2w4C-1iQC-P1k6-UDbh-2skU-TiQtTa

--- Physical volume ---
PV Name               /dev/sde1
VG Name               vg_lab2
PV Size               <2.00 GiB / not usable 3.00 MiB
Allocatable           yes (but full)
PE Size               4.00 MiB
Total PE              511
Free PE               0
Allocated PE          511
PV UUID               cGct7f-KFTN-sFTR-GGwH-Bqhx-k20e-ANWHz
```

```
[03-03-2026 01:45:56] {66381ca2} student@server:~$ sudo vgdisplay
--- Volume group ---
VG Name               vg_lab2
System ID
Format                lvm2
Metadata Areas        3
Metadata Sequence No  4
VG Access              read/write
VG Status              resizable
MAX LV                0
Cur LV               1
Open LV               1
Max PV                0
Cur PV               3
Act PV                3
VG Size               <5.99 GiB
PE Size               4.00 MiB
Total PE              1533
Alloc PE / Size       1533 / <5.99 GiB
Free PE / Size        0 / 0
VG UUID               nBZj3Q-nQ2V-ts1K-D3G1-xV3B-quWi-XsqHq5

[03-03-2026 01:46:00] {66381ca2} student@server:~$ sudo lvdisplay
--- Logical volume ---
LV Path               /dev/vg_lab2/lv_vol01
LV Name               lv_vol01
VG Name               vg_lab2
LV UUID               WzTxdK-9aKk-1lNE-idib-FZ4Q-WBM3-seCwzU
LV Write Access        read/write
LV Creation host, time server, 2026-03-03 00:54:54 +0300
LV Status              available
# open                 1
LV Size               <5.99 GiB
Current LE             1533
Segments               2
Allocation             inherit
Read ahead sectors     auto
- currently set to    512
Block device           252:0
```

20) На server установлен пакет nfs-kernel-server, служба добавлена в автозапуск и запущена

```
[03-03-2026 01:53:04] {66381ca2} student@server:~$ sudo systemctl enable nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nfs-kernel-server
[03-03-2026 01:53:14] {66381ca2} student@server:~$ sudo systemctl start nfs-kernel-server
[03-03-2026 01:53:19] {66381ca2} student@server:~$ sudo systemctl status nfs-kernel-server --no-pager
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: enabled)
   Active: active (exited) since Tue 2026-03-03 01:52:55 MSK; 29s ago
     Main PID: 3150 (code=exited, status=0/SUCCESS)
        CPU: 20ms

Mar 03 01:52:55 server systemd[1]: Starting nfs-server.service - NFS server and services...
Mar 03 01:52:55 server exportfs[3148]: exportfs: can't open /etc/exports for reading
Mar 03 01:52:55 server systemd[1]: Finished nfs-server.service - NFS server and services.
[03-03-2026 01:53:25] {66381ca2} student@server:~$ sudo ss -tulnp | grep 2049
tcp        LISTEN    0      64      0.0.0.0:2049      0.0.0.0:*
tcp        LISTEN    0      64      ::::2049         ::::*
```

21) Настроен экспорт каталога /mnt/vol01 по NFS только для подсети виртуальных машин с правом записи

```
[03-03-2026 01:55:34] {66381ca2} student@server:~$ ip -4 a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 78749sec preferred_lft 78749sec
[03-03-2026 01:59:29] {66381ca2} student@server:~$ echo "/mnt/vol01 10.0.2.0/24(rw, sync, no_subtree_check)" | sudo tee /etc/exports
/mnt/vol01 10.0.2.0/24(rw, sync, no_subtree_check)
[03-03-2026 01:59:33] {66381ca2} student@server:~$ sudo exportfs -ra
[03-03-2026 01:59:41] {66381ca2} student@server:~$ sudo exportfs -v
/mnt/vol01      10.0.2.0/24(sync, wdelay, hide, no_subtree_check, sec=sys, rw, secure, root_squash, no_all_squash)
[03-03-2026 01:59:45] {66381ca2} student@server:~$ sudo systemctl restart nfs-kernel-server
[03-03-2026 01:59:50] {66381ca2} student@server:~$ showmount -e localhost
Export list for localhost:
/mnt/vol01 10.0.2.0/24
```

22) На машине client выполнено монтирование NFS-ресурса в каталог /var/remotenfs

```
[03-03-2026 02:31:34] {734b2653} student@client:~$ showmount -e 192.168.56.10
Export list for 192.168.56.10:
/mnt/vol01 192.168.56.0/24
[03-03-2026 02:34:38] {734b2653} student@client:~$ sudo mount -t nfs 192.168.56.10:/mnt/vol01 /var/remotenfs
[03-03-2026 02:34:44] {734b2653} student@client:~$ mount | grep remotenfs
192.168.56.10:/mnt/vol01 on /var/remotenfs type nfs4 (rw,relatime,vers=4.2,rsize=262144,wsize=262144,namlen=255,hard,proto=tcp,timeo=600,retrans=2,sec=sys,clientaddr=192.168.56.11,local_lock=none,addr=192.168.56.10)
[03-03-2026 02:34:49] {734b2653} student@client:~$ df -h /var/remotenfs
Filesystem                Size      Used Avail Use% Mounted on
192.168.56.10:/mnt/vol01  5.9G         0  5.6G   0% /var/remotenfs
```

23) Успешность монтирования NFS подтверждена, в каталог /var/remotenfs скопирован файл nfs_test.txt

```
[03-03-2026 02:34:44] {734b2653} student@client:~$ mount | grep remotenfs
192.168.56.10:/mnt/vol01 on /var/remotenfs type nfs4 (rw,relatime,vers=4.2,rsize=262144,wsize=262144,namlen=255,hard,proto=tcp,timeo=600,retrans=2,sec=sys,clientaddr=192.168.56.11,local_lock=none,addr=192.168.56.10)
[03-03-2026 02:34:49] {734b2653} student@client:~$ df -h /var/remotenfs
Filesystem                Size      Used Avail Use% Mounted on
192.168.56.10:/mnt/vol01  5.9G         0  5.6G   0% /var/remotenfs
[03-03-2026 02:34:54] {734b2653} student@client:~$ mount | grep /var/remotenfs
192.168.56.10:/mnt/vol01 on /var/remotenfs type nfs4 (rw,relatime,vers=4.2,rsize=262144,wsize=262144,namlen=255,hard,proto=tcp,timeo=600,retrans=2,sec=sys,clientaddr=192.168.56.11,local_lock=none,addr=192.168.56.10)
[03-03-2026 02:39:09] {734b2653} student@client:~$ df -h /var/remotenfs
Filesystem                Size      Used Avail Use% Mounted on
192.168.56.10:/mnt/vol01  5.9G         0  5.6G   0% /var/remotenfs
[03-03-2026 02:39:15] {734b2653} student@client:~$ echo "lab2 nfs test $(date)" > ~/nfs_test.txt
[03-03-2026 02:39:42] {734b2653} student@client:~$ cp ~/nfs_test.txt /var/remotenfs/
[03-03-2026 02:41:35] {734b2653} student@client:~$ ls -l /var/remotenfs/nfs_test.txt
-rw-rw-r-- 1 student student 46 Mar 3 02:41 /var/remotenfs/nfs_test.txt
[03-03-2026 02:41:41] {734b2653} student@client:~$ cat /var/remotenfs/nfs_test.txt
lab2 nfs test Tue Mar 3 02:39:22 AM MSK 2026
```

24) Для файла /mnt/vol01/nfs_test.txt получена информация об inode и логическом размещении на диске

```
[03-03-2026 02:41:24] {460b6960} student@server:~$ ls -li /mnt/vol01/nfs_test.txt
12 -rw-rw-r-- 1 student student 46 Mar 3 02:41 /mnt/vol01/nfs_test.txt
[03-03-2026 02:45:41] {460b6960} student@server:~$ stat /mnt/vol01/nfs_test.txt
  File: /mnt/vol01/nfs_test.txt
  Size: 46          Blocks: 8          IO Block: 4096   regular file
Device: 252,0      Inode: 12           Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/ student)   Gid: ( 1000/ student)
Access: 2026-03-03 02:41:32.630617529 +0300
Modify: 2026-03-03 02:41:32.636620530 +0300
Change: 2026-03-03 02:41:32.636620530 +0300
 Birth: 2026-03-03 02:41:32.624614529 +0300
[03-03-2026 02:45:47] {460b6960} student@server:~$ sudo filefrag -v /mnt/vol01/nfs_test.txt
Filesystem type is: ef53
File size of /mnt/vol01/nfs_test.txt is 46 (1 block of 4096 bytes)
ext:      logical_offset:    physical_offset: length:   expected: flags:
0:        0.             0:        33280..   33280:    1:        last,eof
/mnt/vol01/nfs_test.txt: 1 extent found
```


25) Жёсткая ссылка и исходный файл имеют одинаковый inode и одно физическое размещение. Символическая ссылка имеет другой inode и хранит путь к целевому файлу. При удалении цели становится битой. Различия объясняются тем, что hard link — дополнительное имя того же inode, а symlink - отдельный объект ФС

```
[03-03-2026 02:45:53] {460b6960} student@server:~$ ln /mnt/vol01/nfs_test.txt /mnt/vol01/nfs_test_hardlink
[03-03-2026 02:52:51] {460b6960} student@server:~$ ln -s /mnt/vol01/nfs_test.txt /mnt/vol01/nfs_test_symlink
[03-03-2026 02:52:57] {460b6960} student@server:~$ ls -li /mnt/vol01/nfs_test.txt /mnt/vol01/nfs_test_hardlink /mnt/vol01/nfs_test_symlink
12 -rw-rw-r-- 0 student student 46 Mar 3 02:41 /mnt/vol01/nfs_test.txt
13 lrwxrwxrwx 1 student student 23 Mar 3 02:52 /mnt/vol01/nfs_test_symlink -> /mnt/vol01/nfs_test.txt
12 -rw-rw-r-- 2 student student 46 Mar 3 02:41 /mnt/vol01/nfs_test.txt
[03-03-2026 02:53:12] {460b6960} student@server:~$ stat /mnt/vol01/nfs_test.txt /mnt/vol01/nfs_test_hardlink /mnt/vol01/nfs_test_symlink
  File: /mnt/vol01/nfs_test.txt
  Size: 46          Blocks: 8          IO Block: 4096   regular file
Device: 252:0     Inode: 12          Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ student)   Gid: ( 1000/ student)
Access: 2026-03-03 02:41:32.630617529 +0300
Modify: 2026-03-03 02:41:32.636620530 +0300
Change: 2026-03-03 02:52:51.471878603 +0300
 Birth: 2026-03-03 02:41:32.624614529 +0300
  File: /mnt/vol01/nfs_test_hardlink
  Size: 46          Blocks: 8          IO Block: 4096   regular file
Device: 252:0     Inode: 12          Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ student)   Gid: ( 1000/ student)
Access: 2026-03-03 02:41:32.630617529 +0300
Modify: 2026-03-03 02:41:32.636620530 +0300
Change: 2026-03-03 02:52:51.471878603 +0300
 Birth: 2026-03-03 02:41:32.624614529 +0300
  File: /mnt/vol01/nfs_test_symlink -> /mnt/vol01/nfs_test.txt
  Size: 23          Blocks: 0          IO Block: 4096   symbolic link
Device: 252:0     Inode: 13          Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/ student)   Gid: ( 1000/ student)
Access: 2026-03-03 02:53:13.233754426 +0300
Modify: 2026-03-03 02:52:57.479881191 +0300
Change: 2026-03-03 02:52:57.479881191 +0300
 Birth: 2026-03-03 02:52:57.479881191 +0300
[03-03-2026 02:53:19] {460b6960} student@server:~$ sudo filefrag -v /mnt/vol01/nfs_test.txt
Filesystem type is: ef53
File size of /mnt/vol01/nfs_test.txt is 46 (1 block of 4096 bytes)
ext:   logical_offset:   physical_offset: length:   expected: flags:
0:      0..          0:      33280..      33280:      1:      last,eof
/mnt/vol01/nfs_test.txt: 1 extent found
[03-03-2026 02:53:40] {460b6960} student@server:~$ sudo filefrag -v /mnt/vol01/nfs_test_hardlink
Filesystem type is: ef53
File size of /mnt/vol01/nfs_test_hardlink is 46 (1 block of 4096 bytes)
ext:   logical_offset:   physical_offset: length:   expected: flags:
0:      0..          0:      33280..      33280:      1:      last,eof
/mnt/vol01/nfs_test_hardlink: 1 extent found
[03-03-2026 02:53:46] {460b6960} student@server:~$ sudo filefrag -v /mnt/vol01/nfs_test_symlink
Filesystem type is: ef53
File size of /mnt/vol01/nfs_test_symlink is 46 (1 block of 4096 bytes)
ext:   logical_offset:   physical_offset: length:   expected: flags:
0:      0..          0:      33280..      33280:      1:      last,eof
/mnt/vol01/nfs_test_symlink: 1 extent found
```